Imperial College London

21 November 2024

Advanced Imaging Technologies in Systems Biology and Biomedical Research

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What we covered

- How to visualise molecular dynamics using fluorescence imaging II?
- What are the basic principles of development?
- How is tissue patterned?

Modes of dynamic fluorescence imaging

FRAP Fluorescence recovery after photobleaching.
FLAP Fluorescence localisation after photobleaching.
FLIP Fluorescence loss in photobleaching.
FDAP Fluorescence decay after photoactivation.

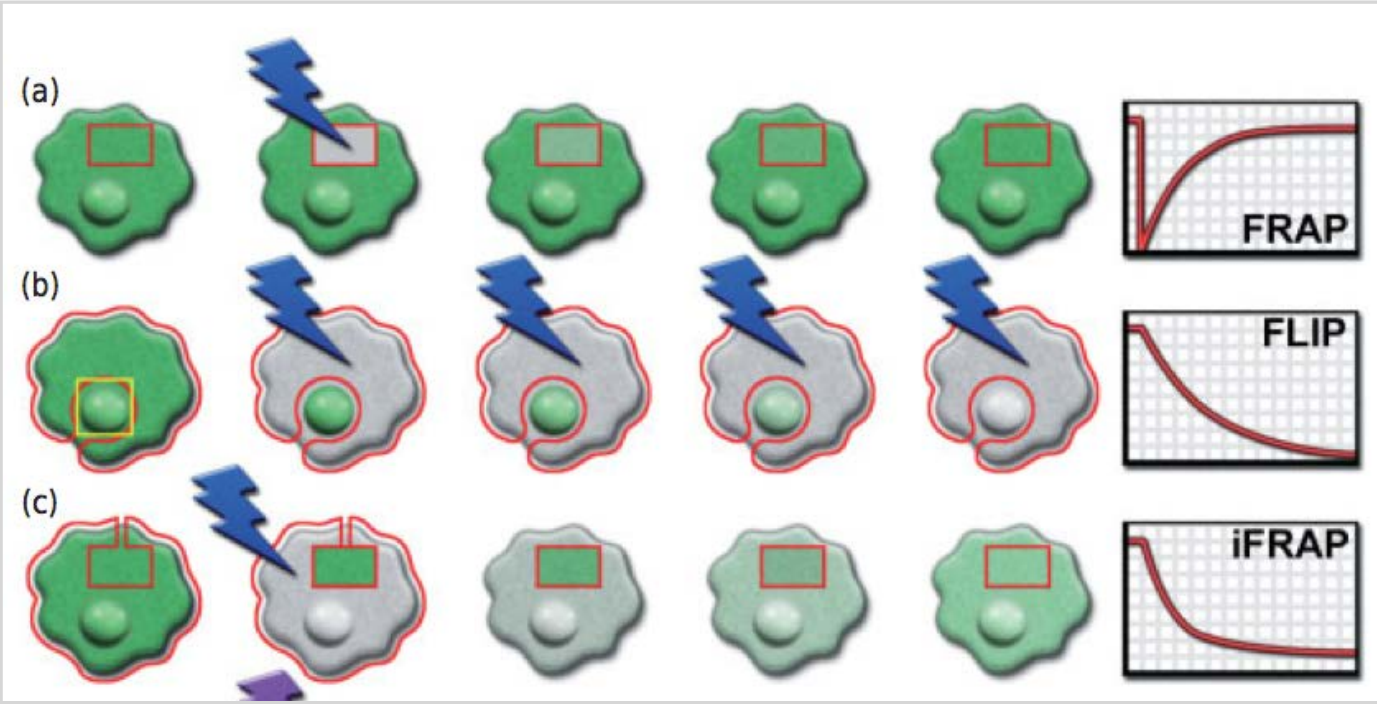
4D/5D

*Time lapse microscopy
in x, y, and z
spatial dimensions in time (4D imaging)
and in multiple colour channels (5D imaging).*

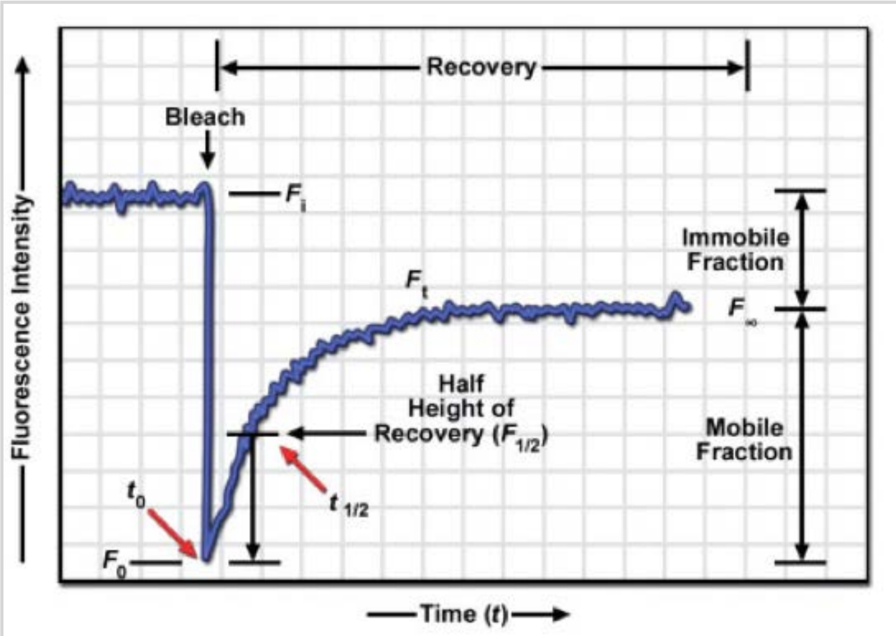
FCS Fluorescence correlation spectroscopy.
FCCS Fluorescence cross-correlation spectroscopy.
ICS Image correlation spectroscopy.

FRET Förster (fluorescence) resonance energy transfer.
BRET Bioluminescence resonance energy transfer.
BIFC Bimolecular Bimolecular fluorescence complementation.
FLIM Fluorescence lifetime imaging microscopy.

FRAP: Techniques used and related imaging modalities



FRAP kinetics



The **mobile fraction**, the fraction of molecules that does show recovery, is calculated from the curve as:

(F_∞/F_i),

the balance is the **immobile fraction**:

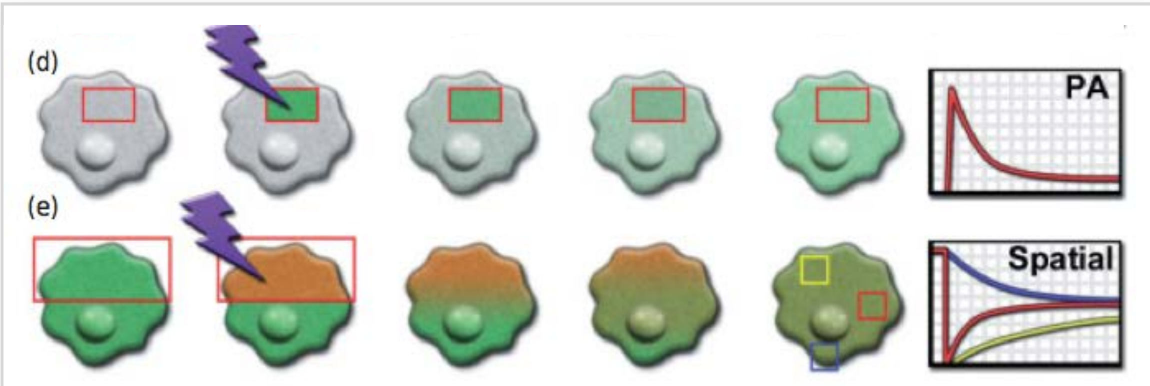
($F_i - F_\infty$)/F_i.

The **percent recovery**, the percent of the initial fluorescence that is recovered, is:

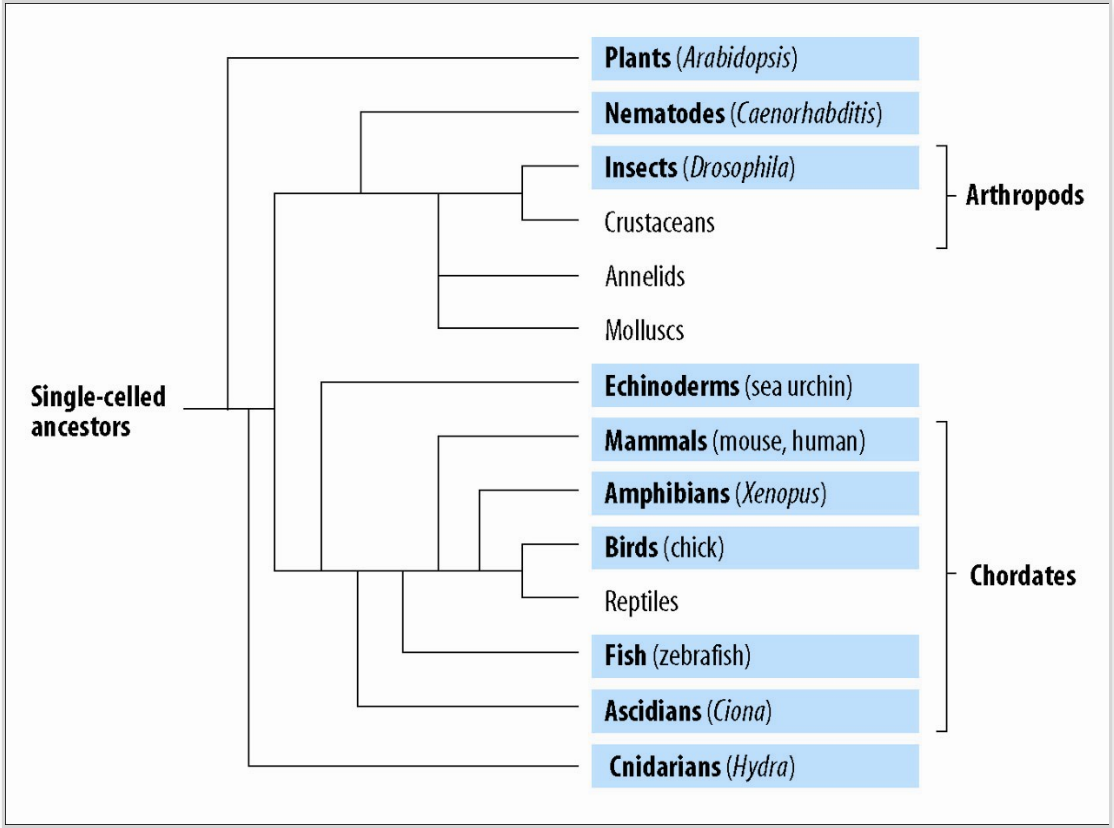
%Recovery = ($F_\infty - F_0$)/($F_i - F_0$),

and ***F_0*** is the fluorescence intensity immediately after bleaching.

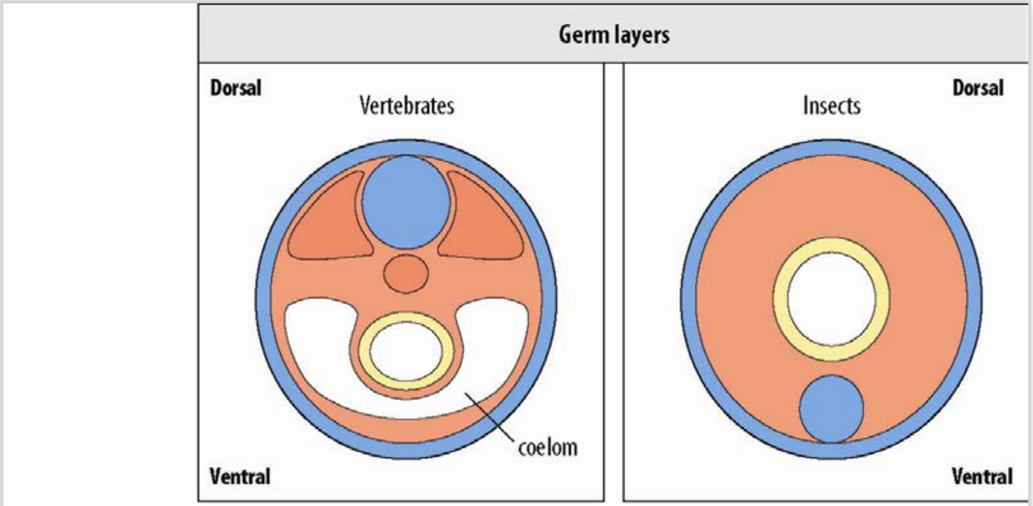
FDAP :
Fluorescence decay after photoactivation/-conversion



Development is studied mainly through selected model organisms

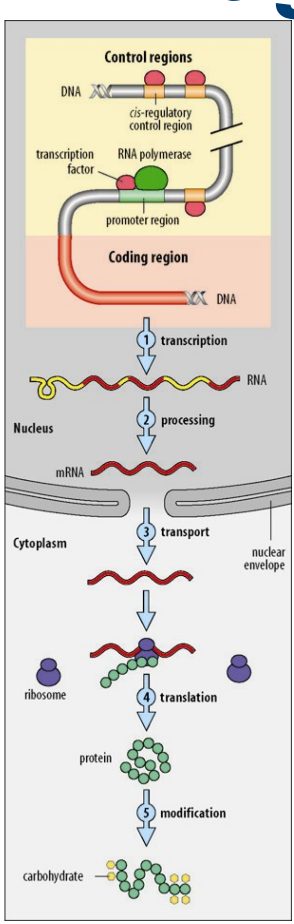


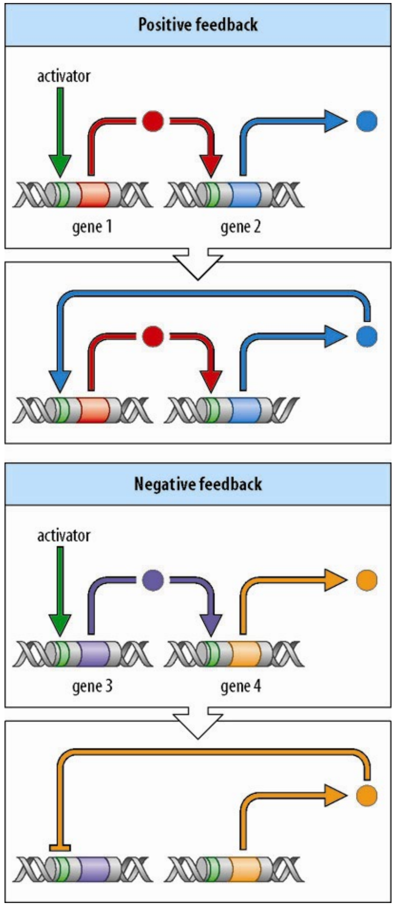
Development involves pattern formation, morphogenesis, cell differentiation and growth… and cells are allocated to different germ layers



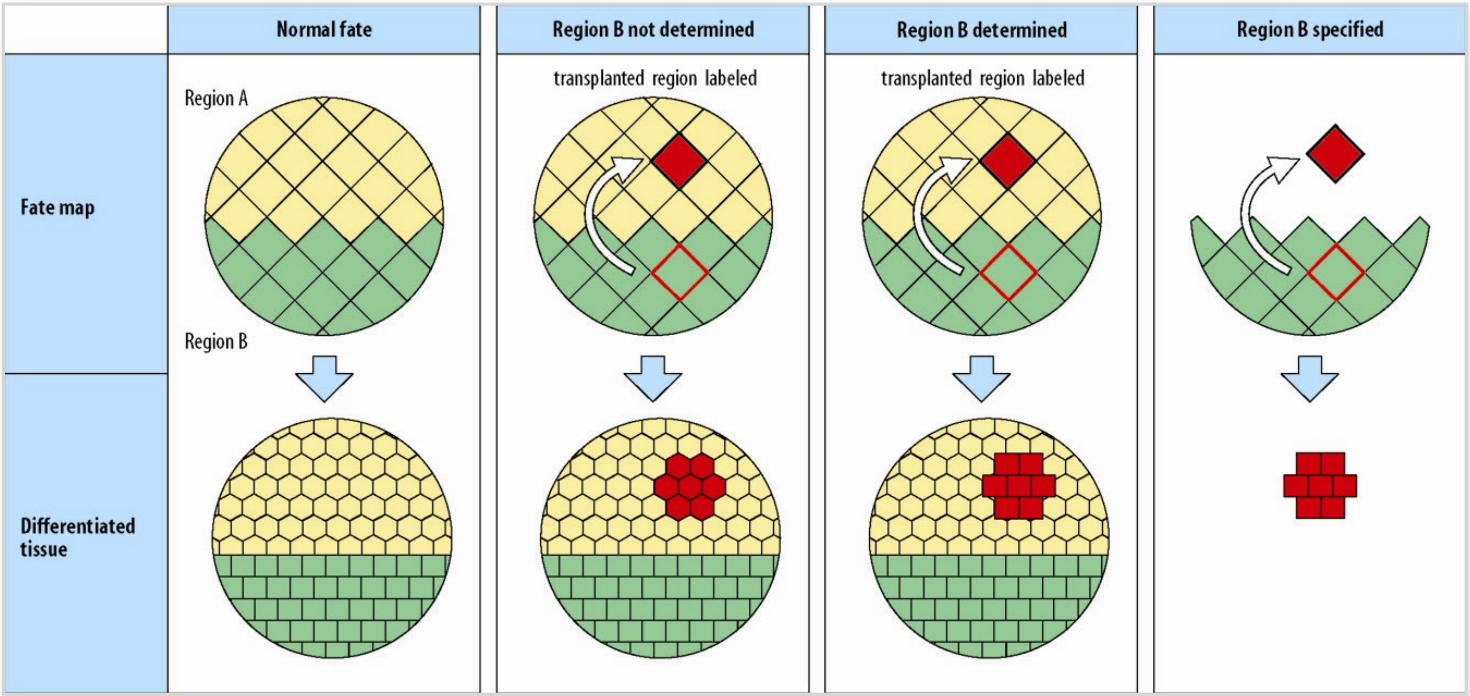
Germ layers	Organs	
Endoderm	gut, liver, lungs	gut
Mesoderm	skeleton, muscle, kidney, heart, blood	muscle, heart, blood
Ectoderm	epidermis of skin, nervous system	cuticle, nervous system

The expression of developmental genes is under tight control

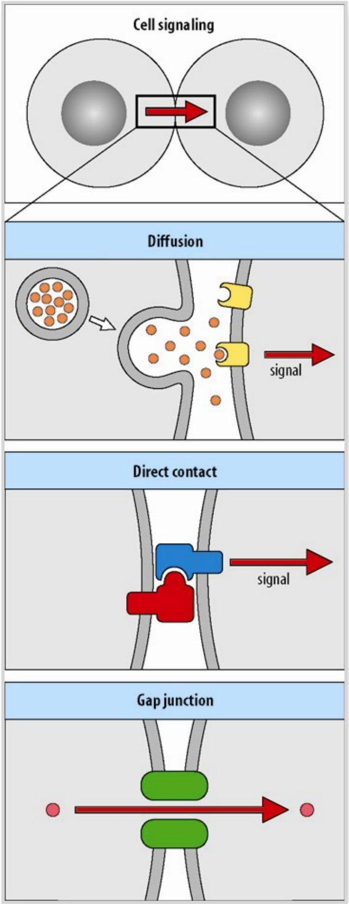




Development is progressive and the fate of cells becomes determined at different times



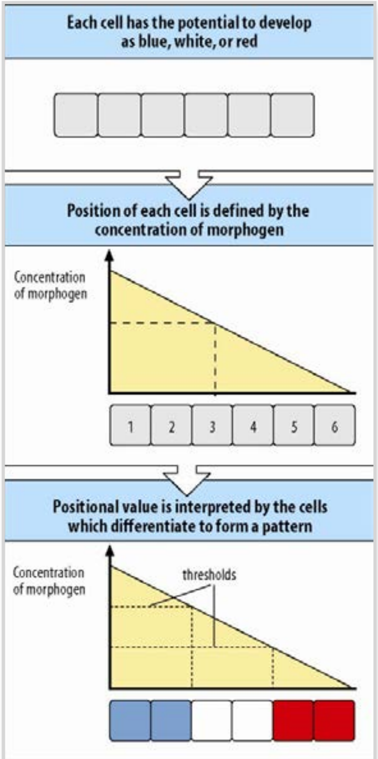
Inductive interactions can make cells different from each other

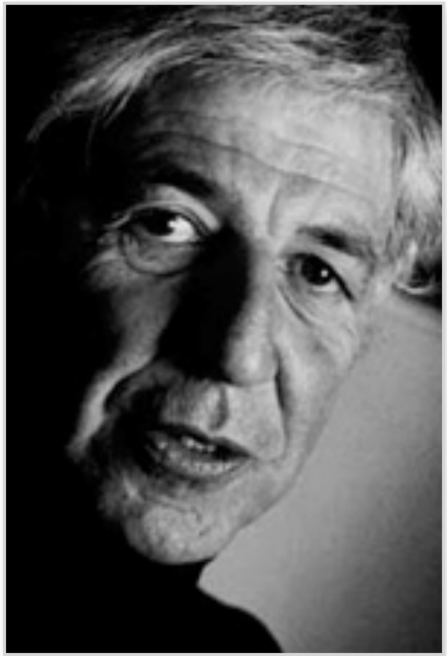


permissive induction:
- one kind of response
- depends on signal threshold

instructive induction:
- different responses to different concentrations

Patterning can involve the interpretation of positional information





Lewis Wolpert (b.1929)

1) positional information has to be related to some boundary
2) this positional information has then to be interpreted
(Note: no need to set relation between positional values and how they are interpreted)

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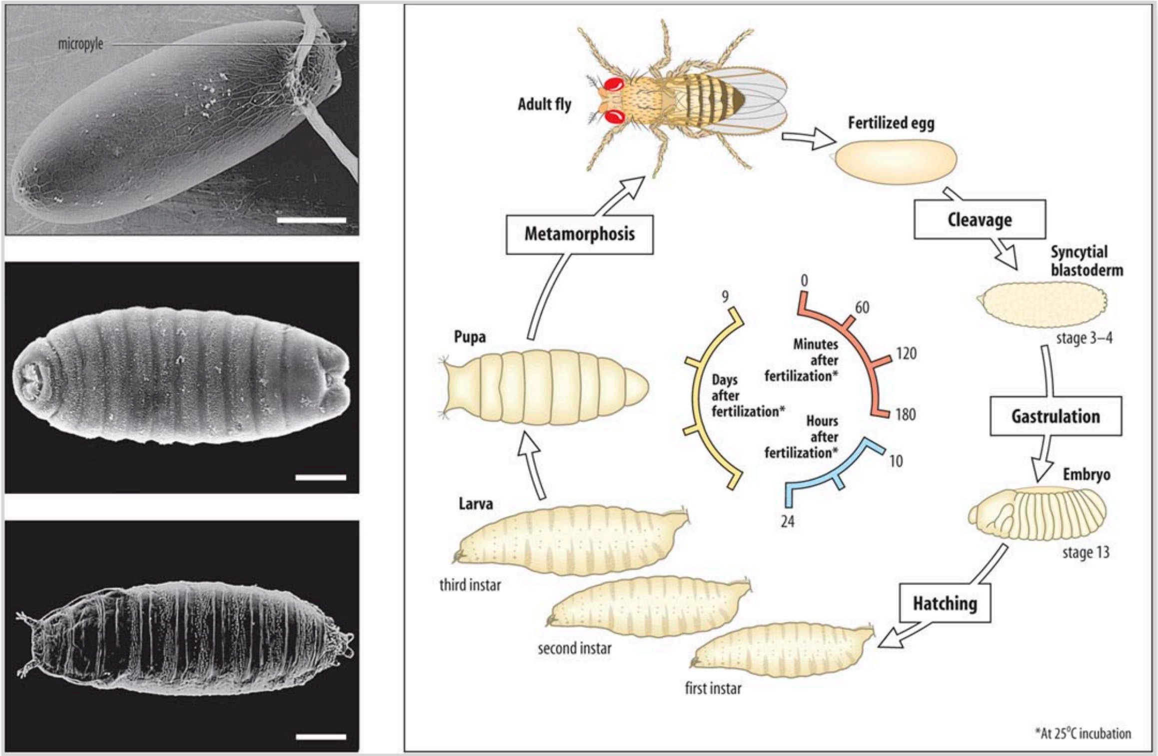
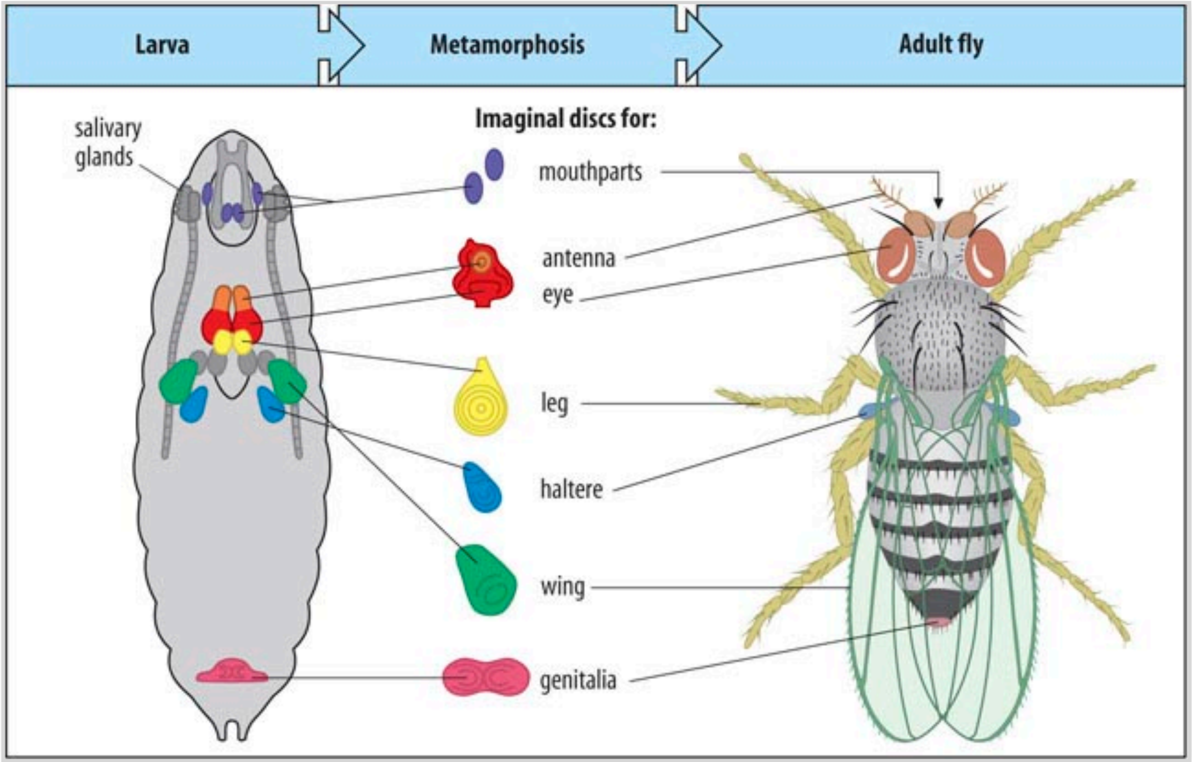
- How to visualise molecular dynamics using fluorescence imaging?
- What are the basic principles of development?
- **How is tissue patterned?**

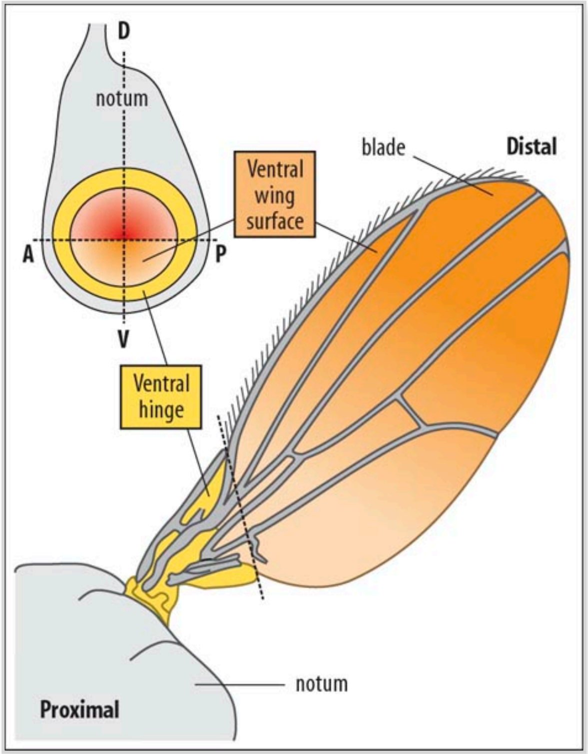
Drosophila melanogaster life cycle and overall development



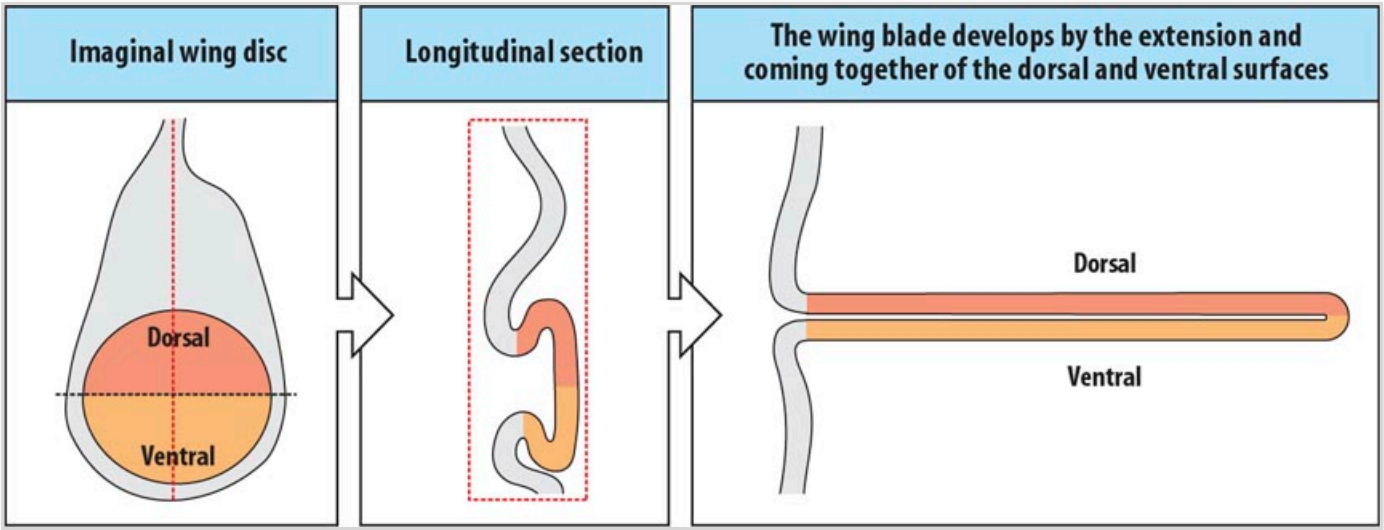
Imaginal discs give rise to adult structures at metamorphosis



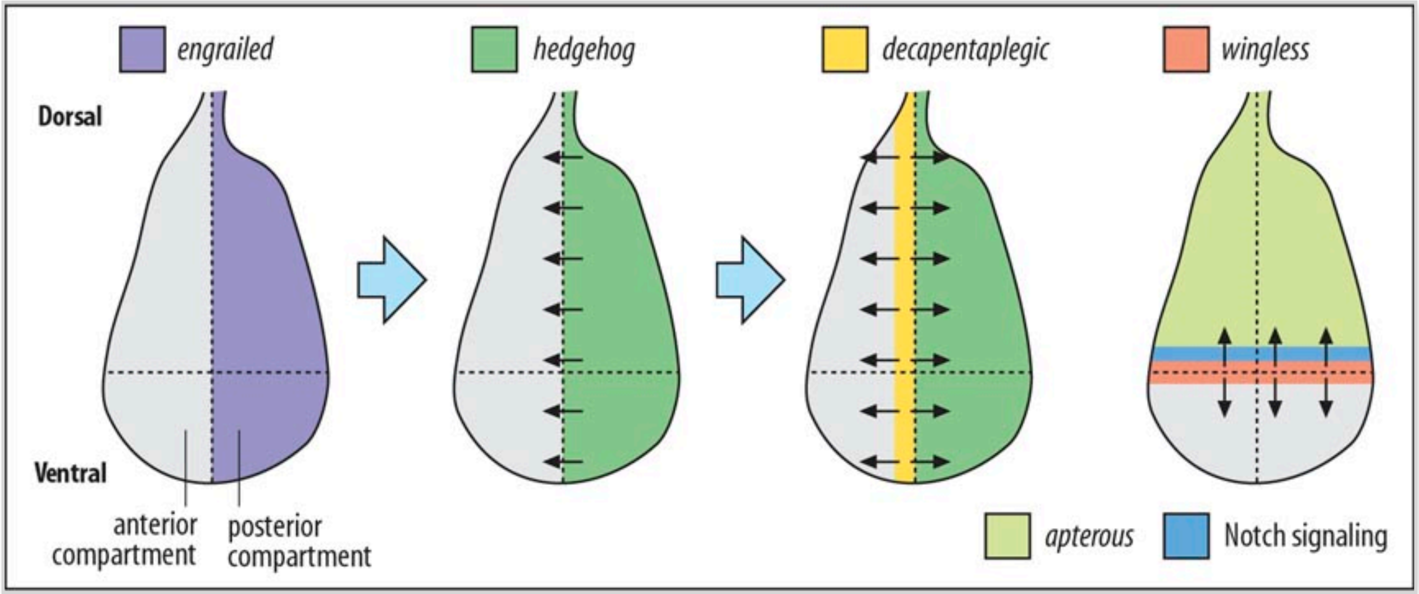
Fate map of the wing imaginal disc of drosophila melanogaster



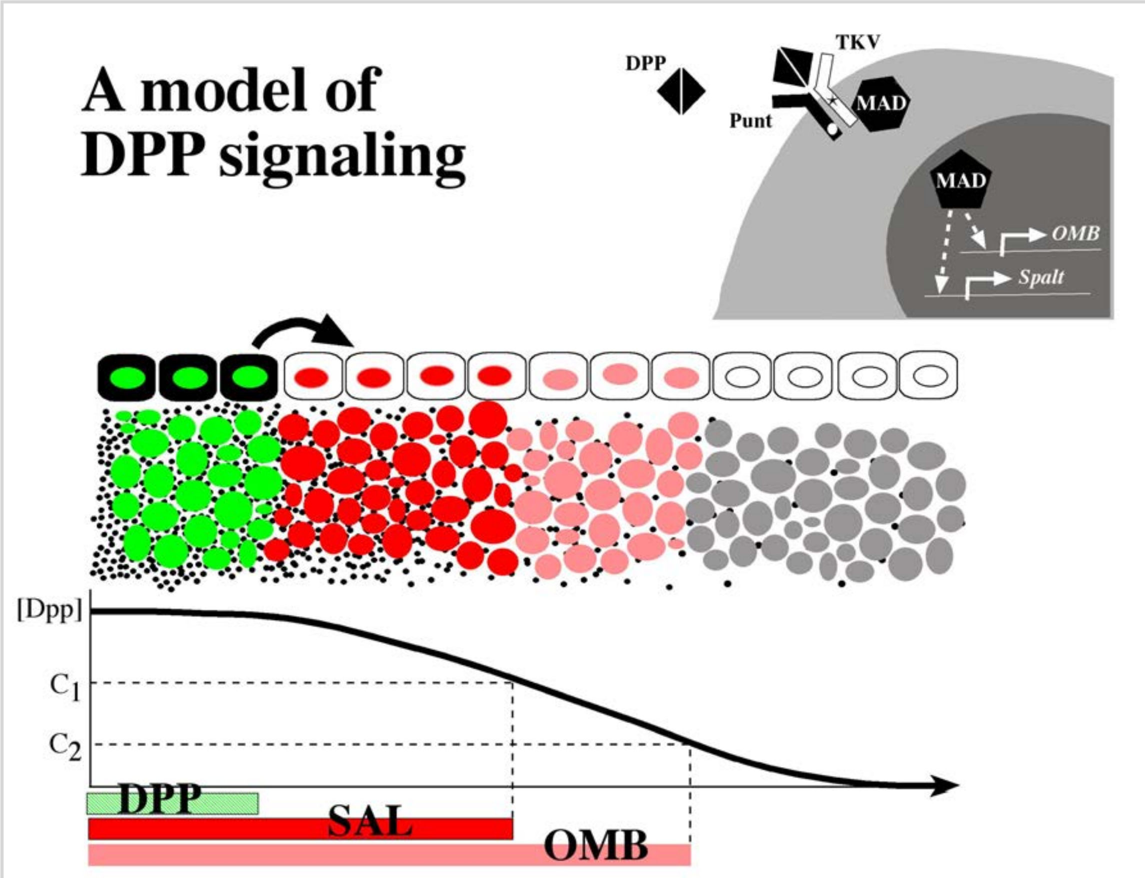
Development of the wing blade from the imaginal disc



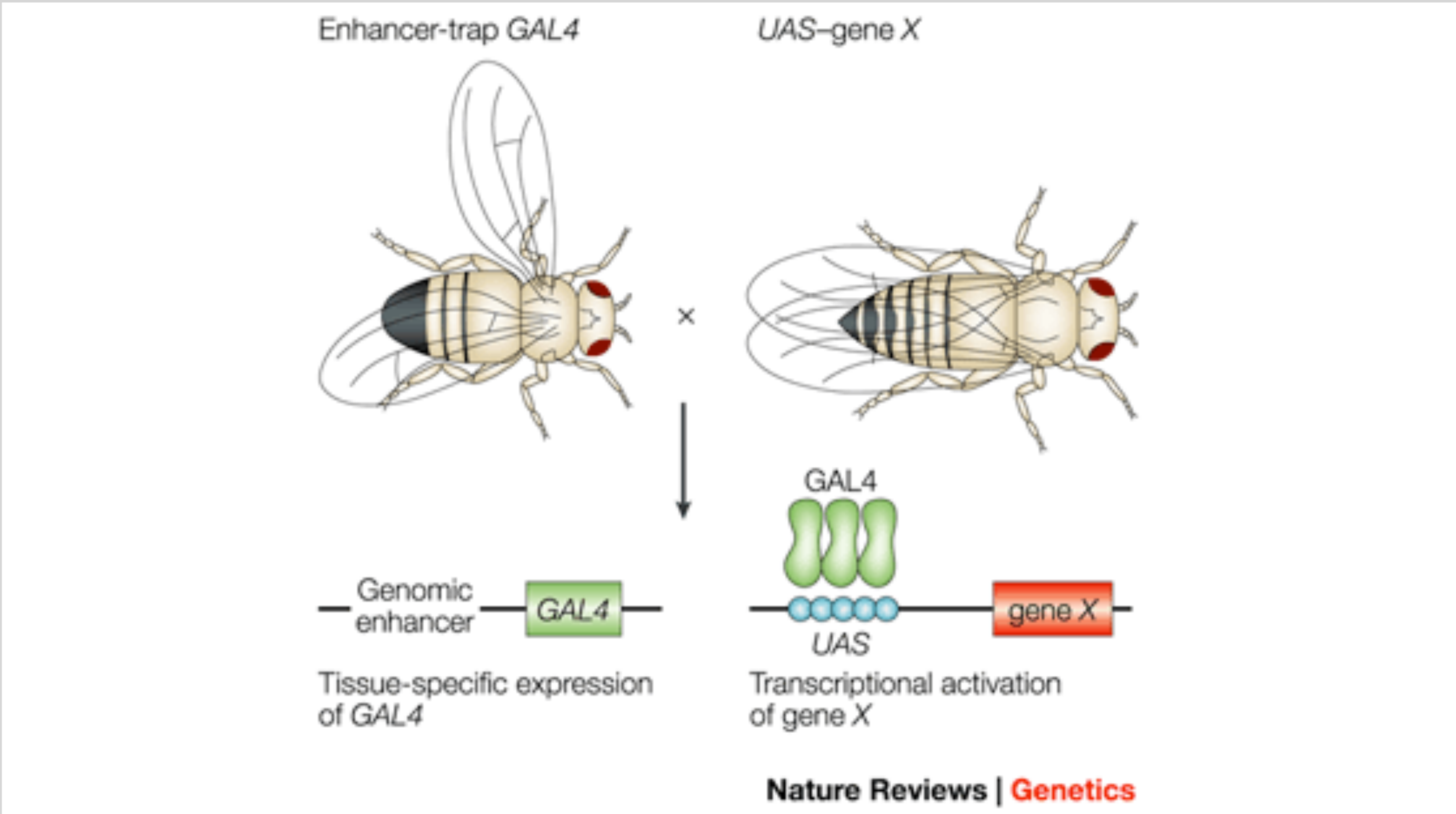
Establishment of signalling regions in the wing disc at the compartment boundaries



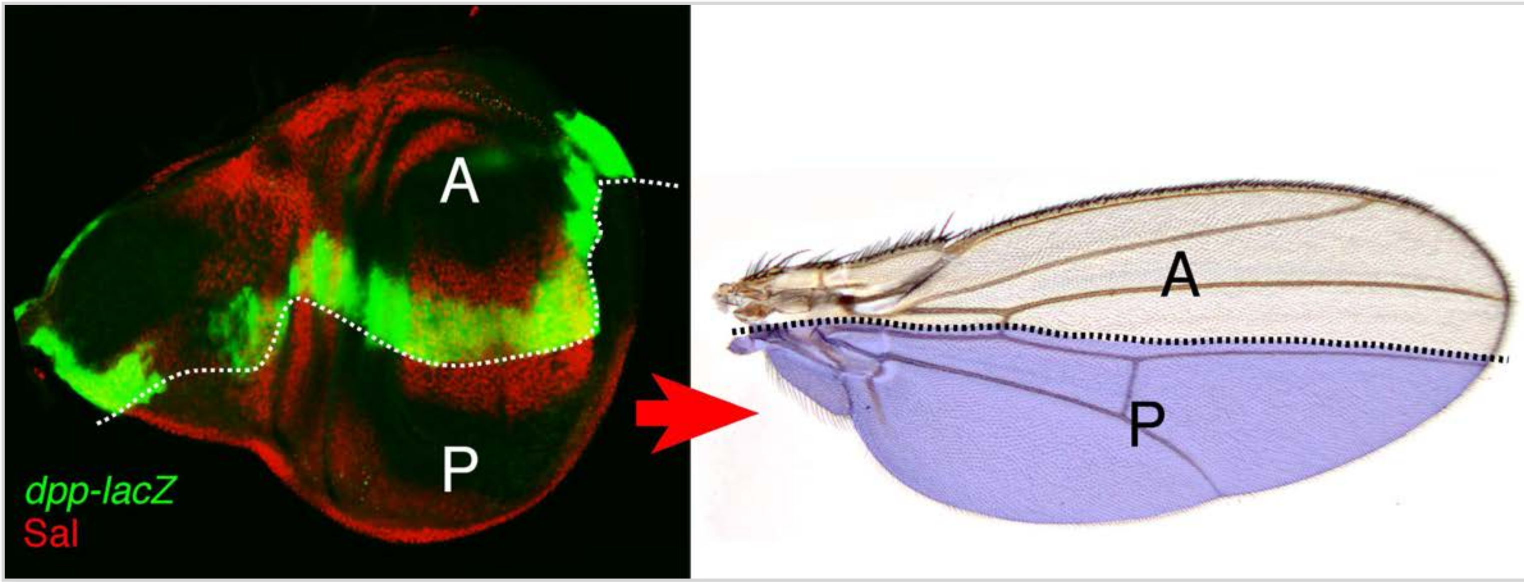
Dpp is a long range morphogen



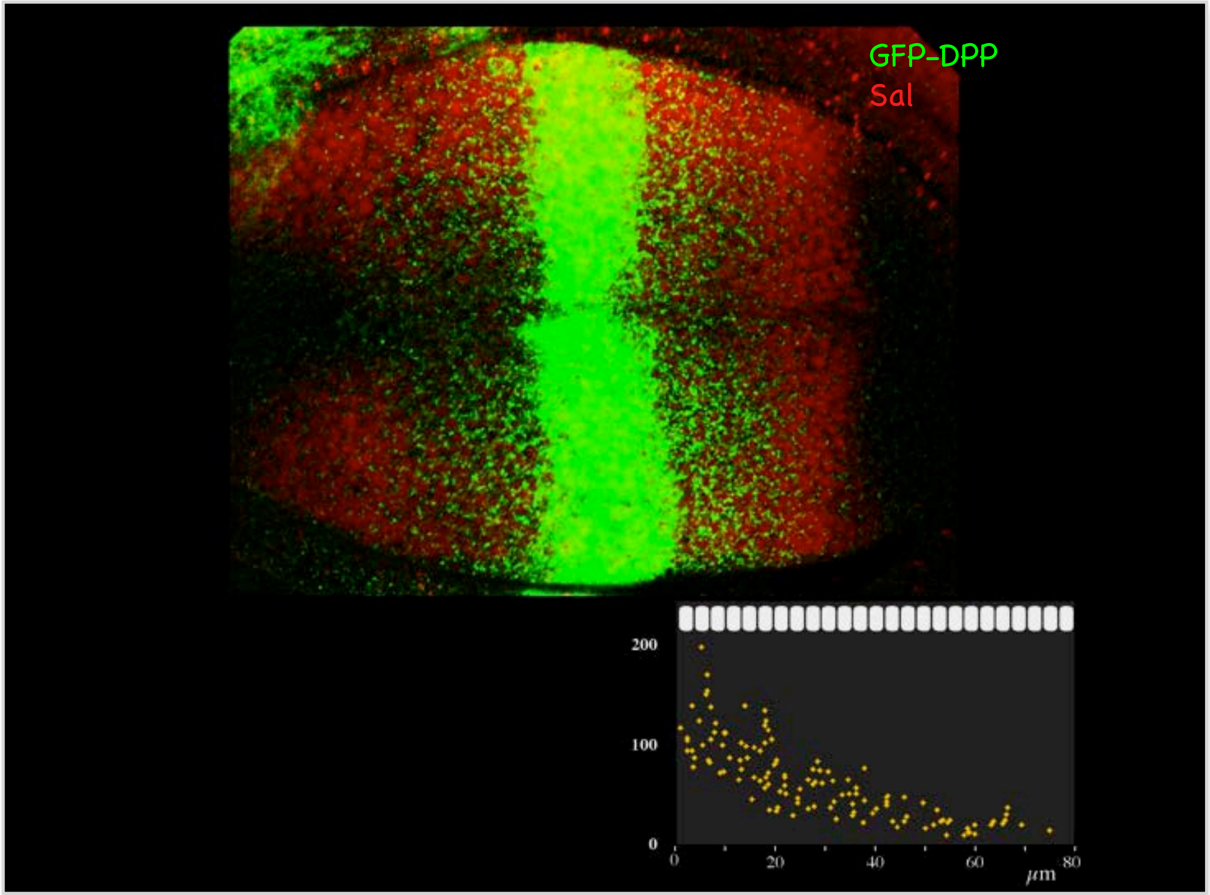
The GAL4–UAS system for directed gene expression



DPP signalling in the wing



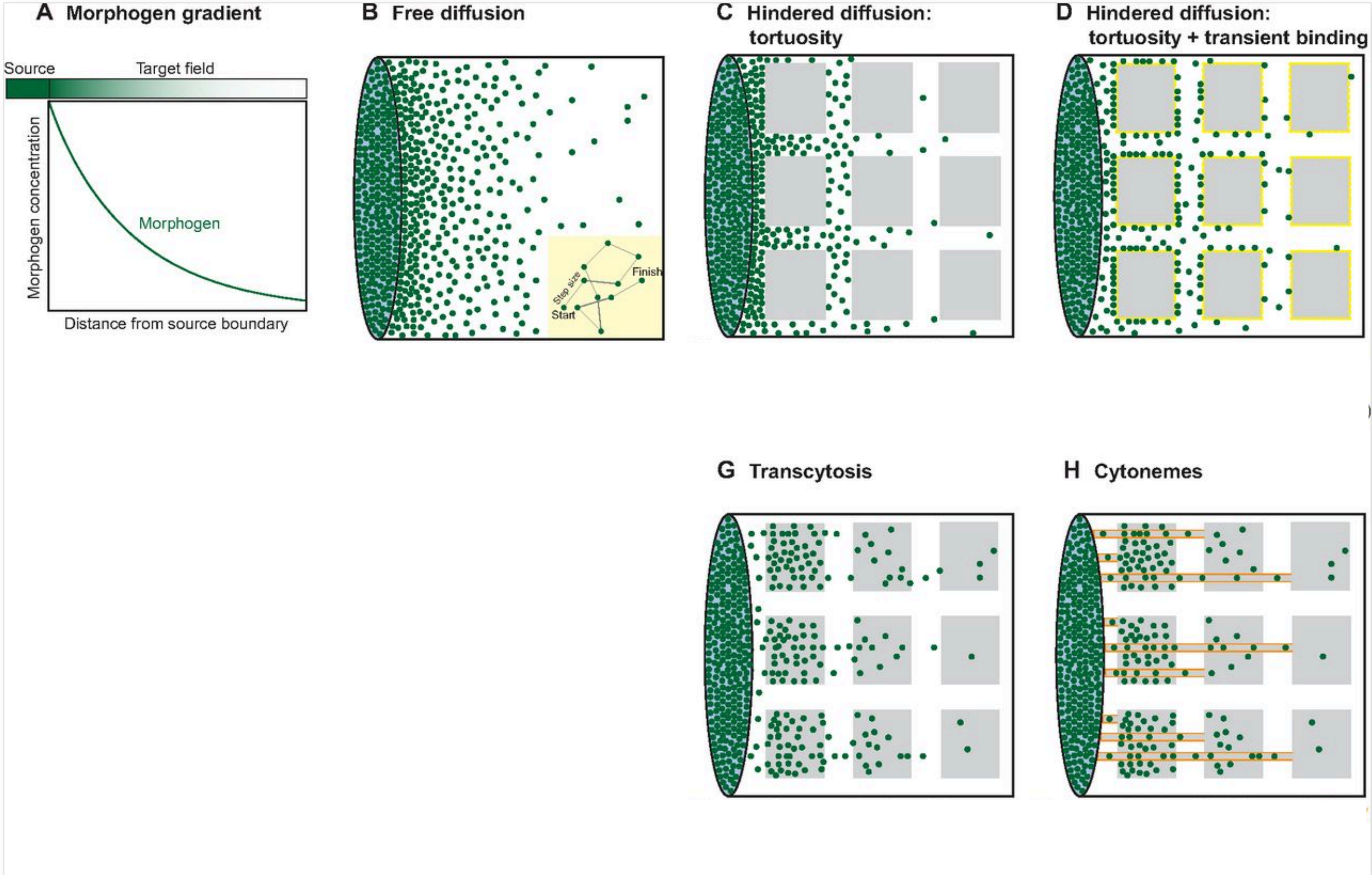
DPP is secreted and distributed as a gradient



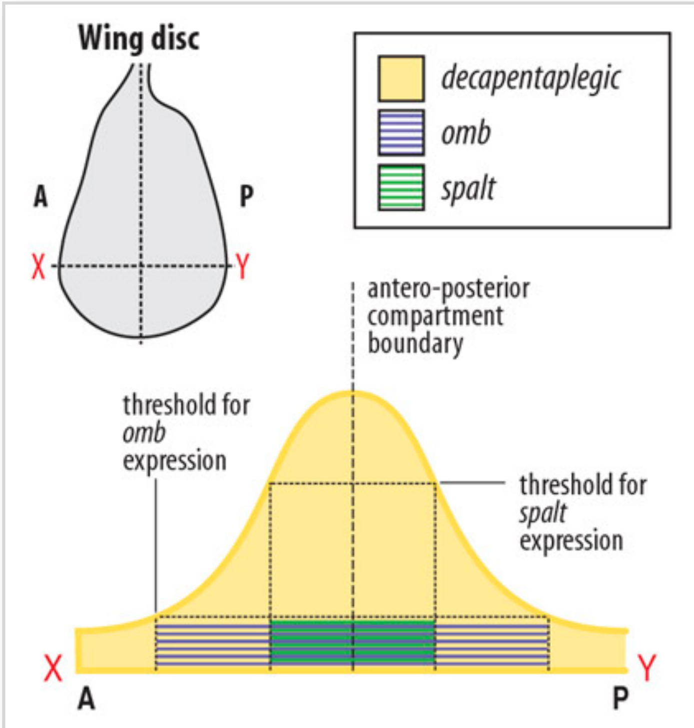
What we will cover

- **Quantitative imaging of morphogen gradients in Drosophila melanogaster**
- Diffusion or planar transcytosis?

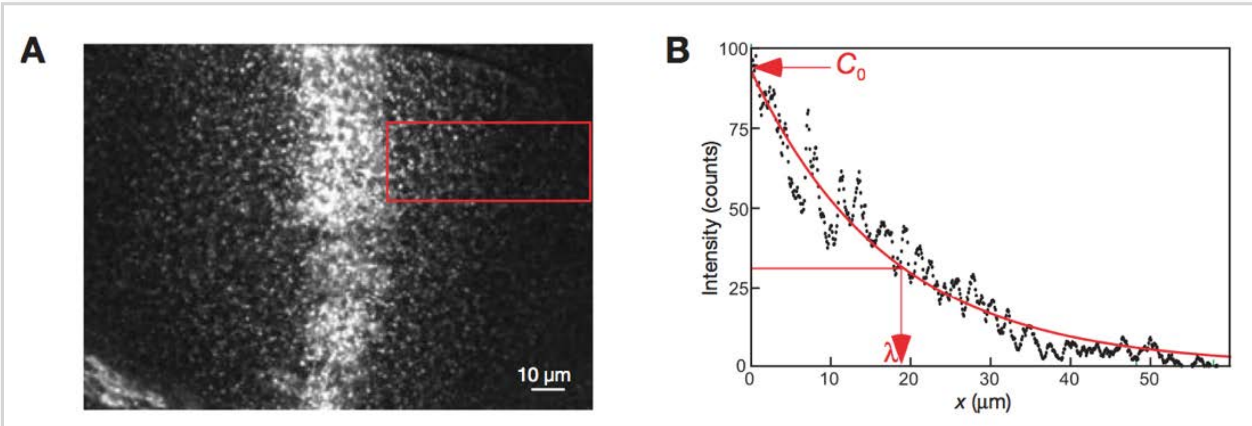
Morphogen transport models



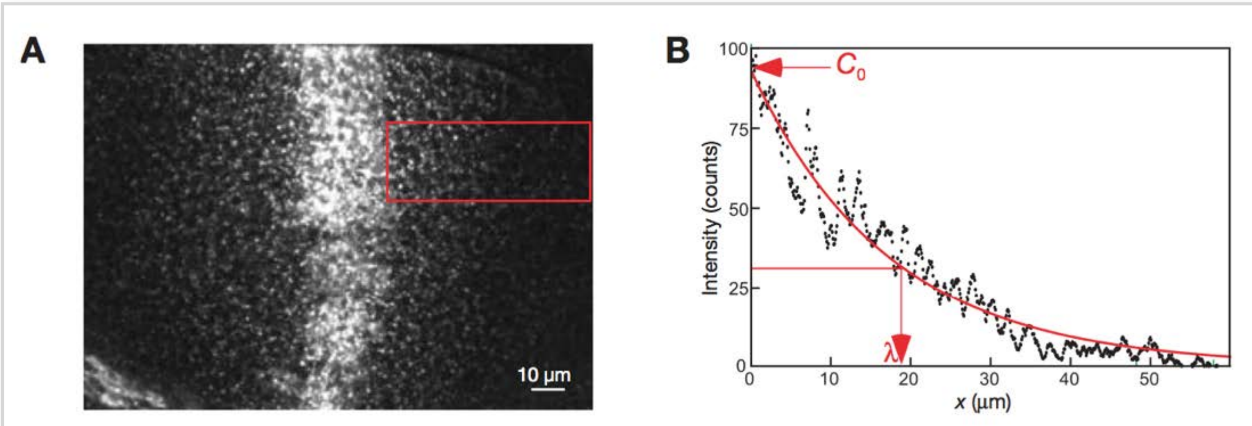
Tissue patterning by morphogen gradients



Measuring morphogen gradient shape



Measuring morphogen gradient shape

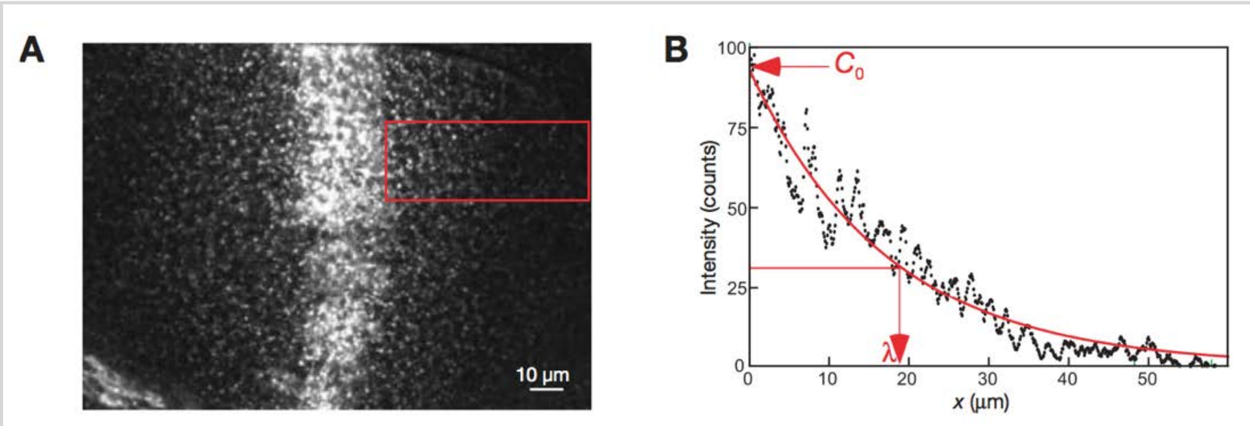


Gradient shape can be more generally described by a theoretical function

Situation

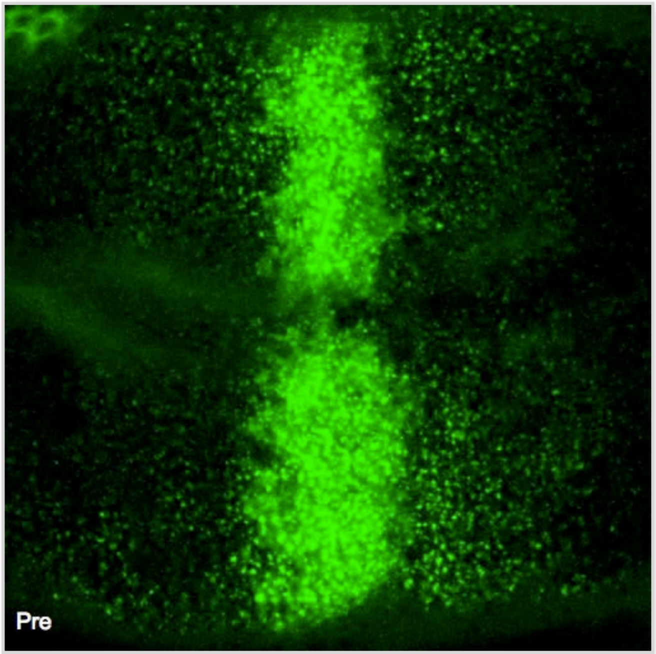
morphogens that are secreted from a localised source, spread non-directionally, and are uniformly degraded in the tissue form exponential gradients

Quantifying morphogen gradient shape

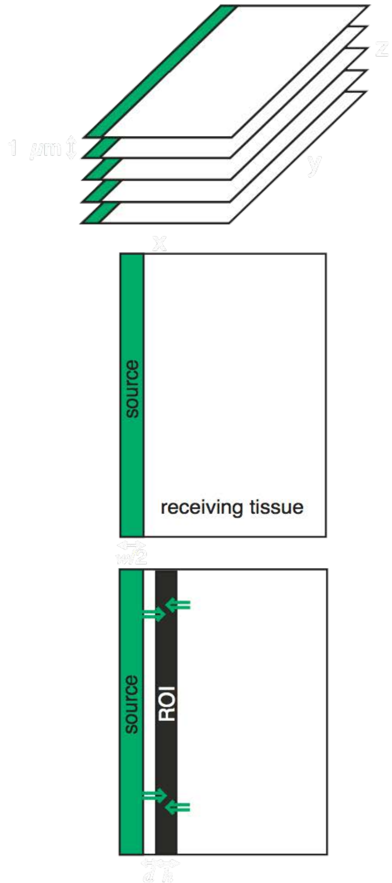


Fluorescence recovery after photobleaching (FRAP): kinetics of morphogen gradient formation

FRAP of GFP-Dpp



dppGal4::UAS-GFP-Dpp/+

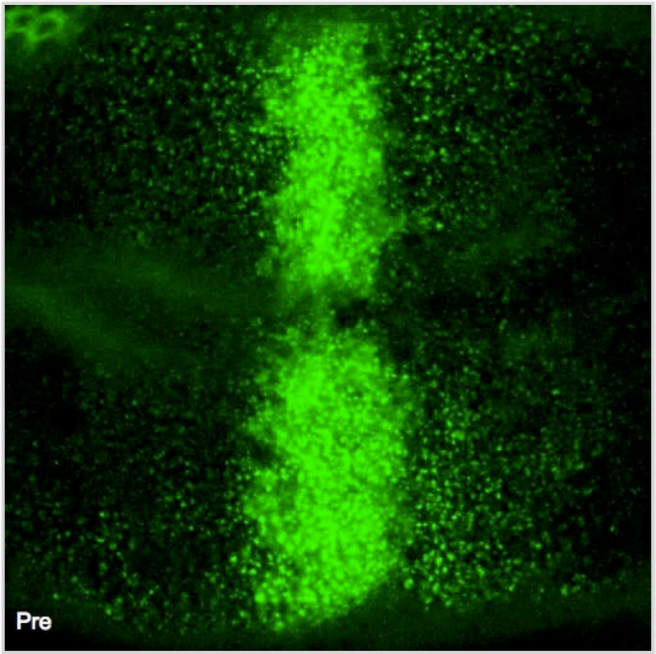


- 3D z-stack
- 2D Projection
- 1D Recovery

Science. 2007 Jan 26;315(5811):521-5.

Fluorescence recovery after photobleaching (FRAP): kinetics of morphogen gradient formation

FRAP of GFP-Dpp



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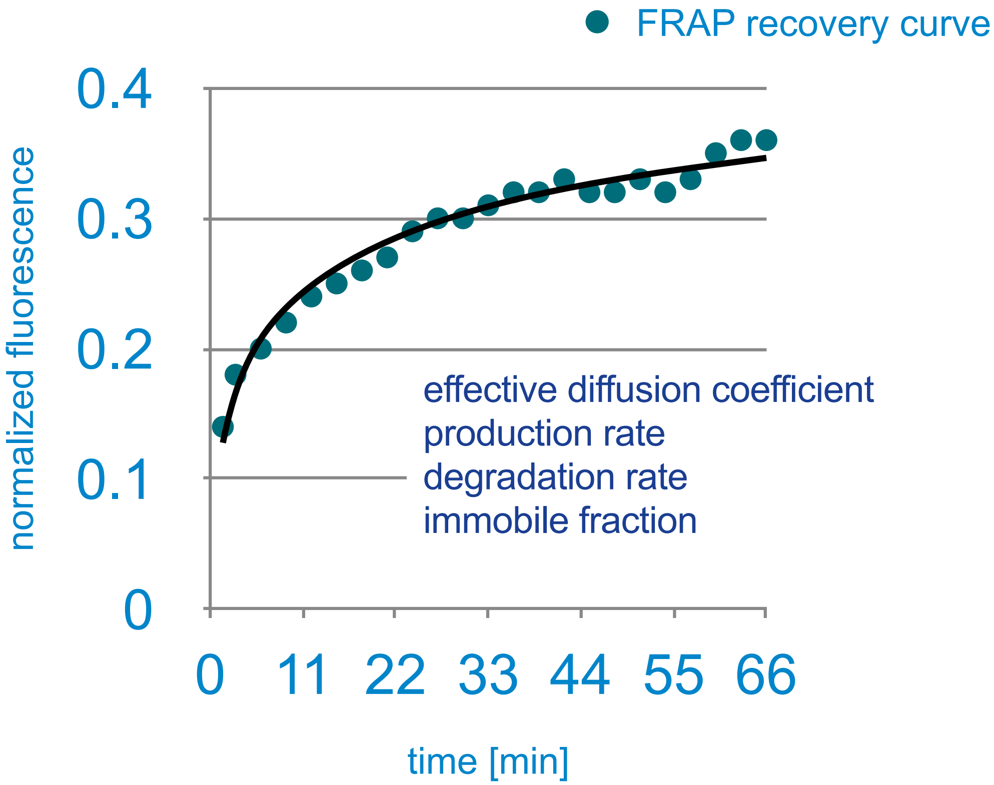
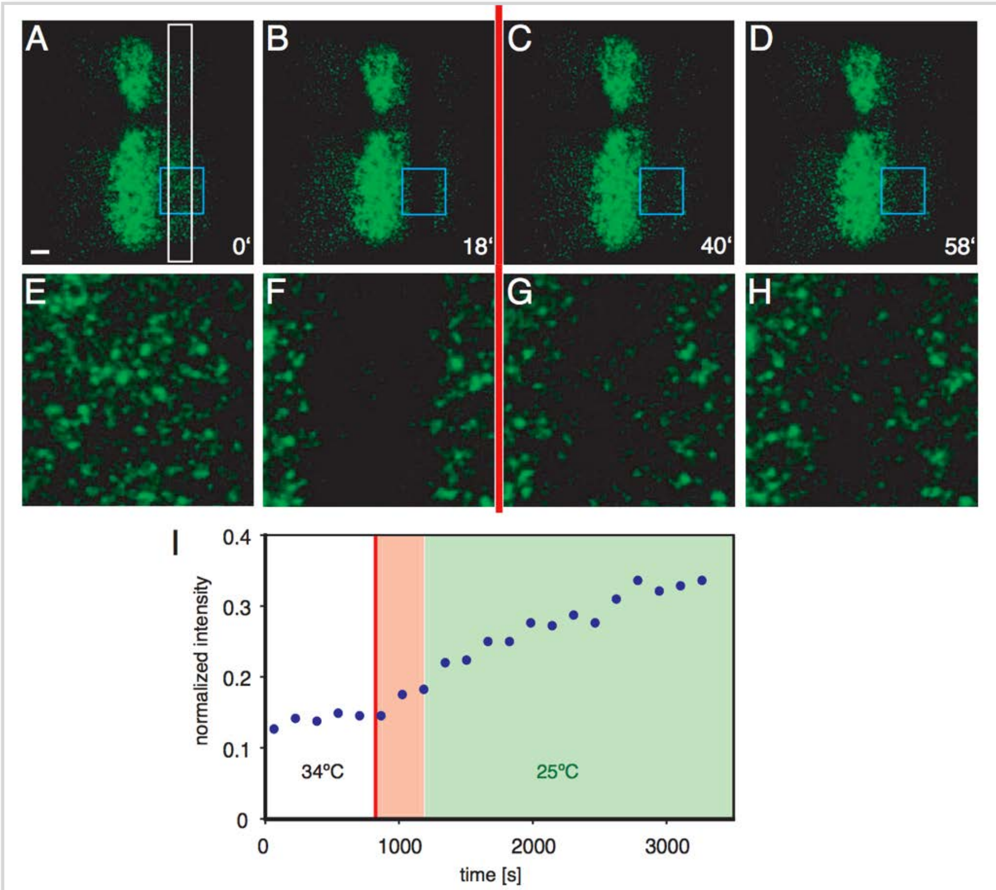


Science. 2007 Jan 26;315(5811):521-5.

The block of endocytosis in the “shibire rescue” experiments is reversible



What we will cover

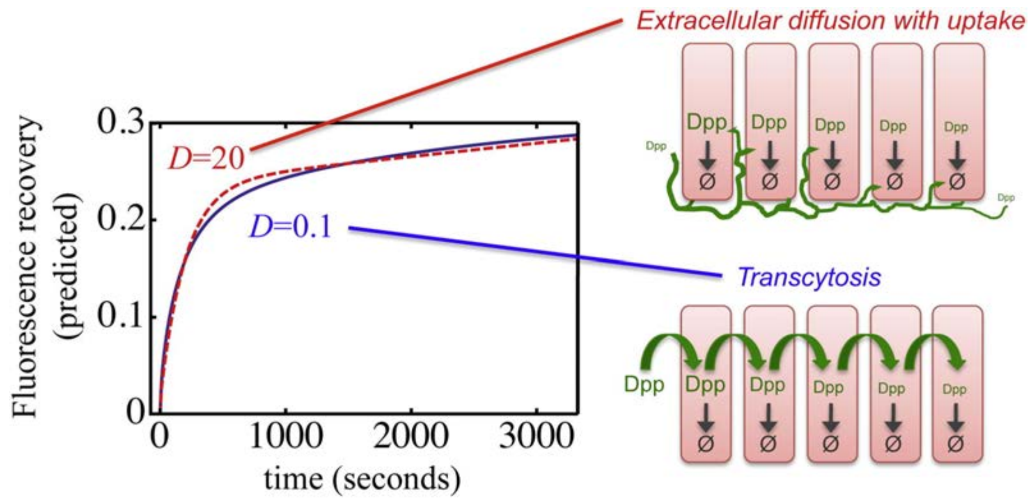
- Quantitative imaging of morphogen gradients in Drosophila melanogaster
- **Diffusion or planar transcytosis?**

What mechanism creates the Dpp morphogen gradient of the Drosophila wing disc

How does the Dpp Morphogen gradient form?

→ Transcytosis?
- FRAP experiments by Kicheva et. al and Gregor et. al.
- Low Diffusion coeff. (<1 μm2/s)
- Large Dpp fraction diffusing

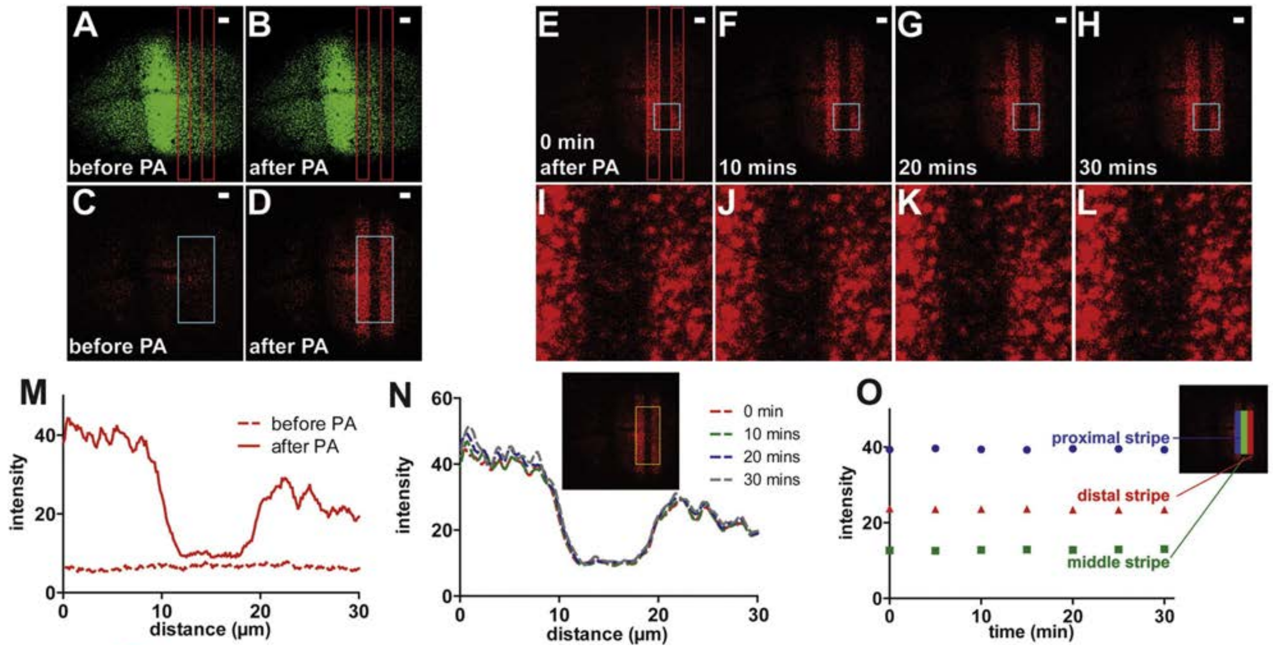
→ Extracellular diffusion?
- Fast, unhindered diffusion
- High Diffusion coeff. (>20 μm2/s)
- High receptor-mediated uptake and degradation



What mechanism creates the Dpp morphogen gradient of the Drosophila wing disc

- Dendra2 -> green to red switch

- No significant spreading of red fluorescence

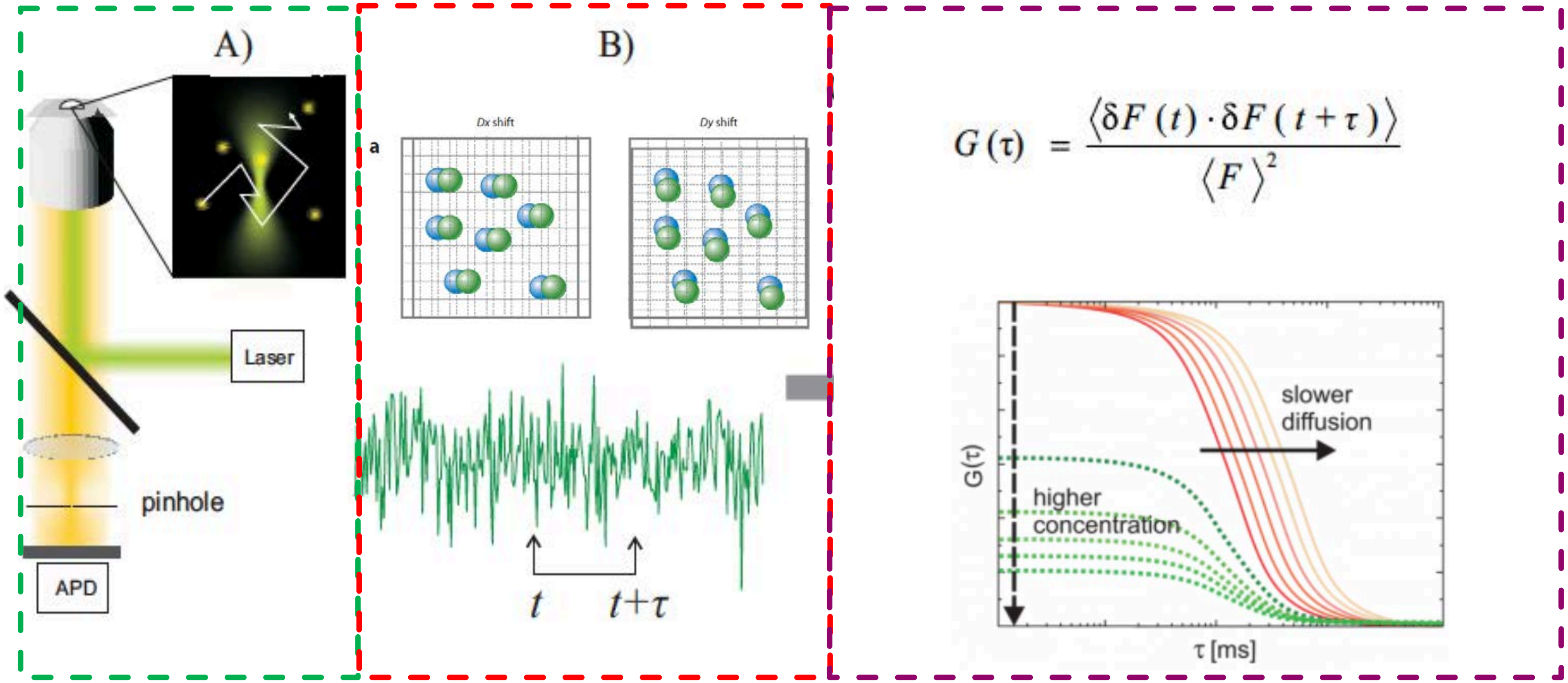


General principle of FCS

1. Confocal microscopy

2. Concentration & intensity fluctuation

3. Time-correlation analysis



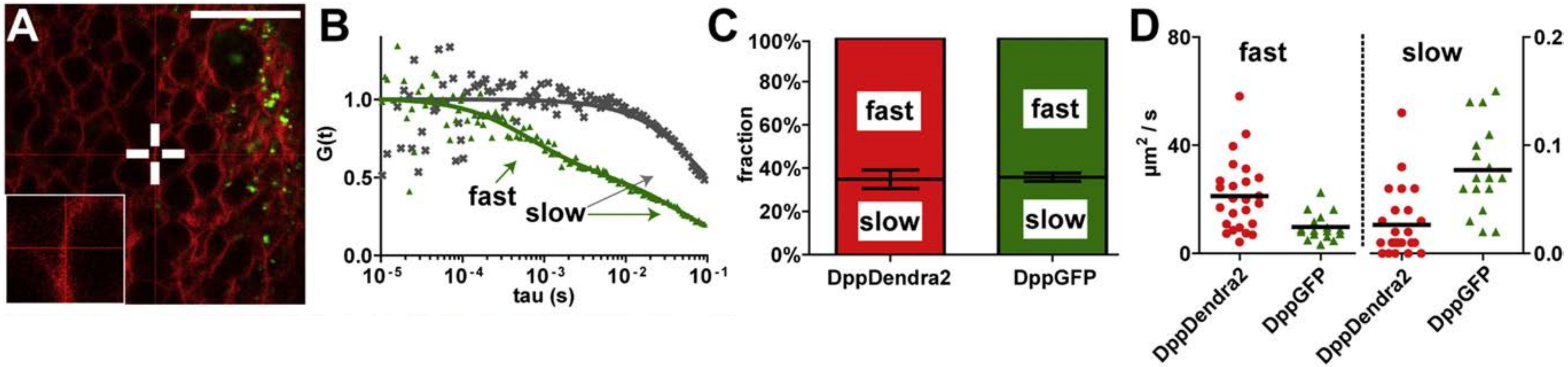
Fluorescence correlation spectroscopy (FCS)

- Fluorescence correlation spectroscopy (FCS) is a powerful technique to measure **concentrations, mobilities, and interactions of fluorescent biomolecules**

- It can be applied to various biological systems such as simple homogeneous solutions, cells, artificial, or cellular membranes and whole organisms

- FCS measurements are based on statistical analysis of fluorescence intensity **fluctuations** from fluorescently labelled molecules inside small confocal volume

Two species of Dpp kinetics



- Two species:

- Free moving DppDendra2 (65%)
    - D = 21 µm2/s

- Membrane bound DppDendra2 (35%)
    - D << 1 µm2/s

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Thank you!

Periklis (Laki) Pantazis